

# RAGAS vs. DeepEval — Final Evaluation Report

Track B — Phase 1: Evaluation-Framework Comparison

July 23, 2026

## 1 Purpose

This report synthesizes the framework comparison between RAGAS (Es et al., 2023) and DeepEval based on empirical execution metrics and evaluation outputs. The analysis assesses framework performance across precision, recall, faithfulness, and answer relevancy metrics using the project’s self-hosted retrieval and generation pipeline.

## 2 Run Configuration

| Field              | Value                                                         |
|--------------------|---------------------------------------------------------------|
| Judge model        | qwen2.5:3b (Ollama, self-hosted)                              |
| Embedding model    | all-MiniLM-L6-v2                                              |
| Case count         | 7 test queries (from <code>queries/test_queries.json</code> ) |
| Retrieval top-k    | 3                                                             |
| Run date           | July 23, 2026                                                 |
| Hardware execution | CPU/GPU shared pipeline via <code>ollama ps</code>            |

## 3 Results

| Framework | Metric               | Ref. req’d | Score  | Case count | Time (s) |
|-----------|----------------------|------------|--------|------------|----------|
| RAGAS     | faithfulness         | No         | 0.8214 | 7          | 42.31    |
| RAGAS     | answer_relevancy     | No         | 0.7689 | 7          | 38.77    |
| RAGAS     | context_precision    | Yes        | 0.7143 | 7          | 45.02    |
| RAGAS     | context_recall       | Yes        | 0.6857 | 7          | 44.88    |
| DeepEval  | faithfulness         | No         | 0.8571 | 7          | 301.44   |
| DeepEval  | answer_relevancy     | No         | 0.7429 | 7          | 287.19   |
| DeepEval  | contextual_precision | Yes        | 0.6786 | 7          | 295.63   |
| DeepEval  | contextual_recall    | Yes        | 0.6571 | 7          | 278.90   |

Table 2: Empirical framework execution summary compiled from `results_eval_frameworks.csv`.

## 4 Analysis and Observations

### 4.1 Score Comparison

The evaluation frameworks exhibit consistent alignment across the reference-free metrics (**faithfulness** and **answer\_relevancy**), varying by only 0.02–0.04. This convergence confirms both frameworks evaluate straightforward queries similarly, with slight variations arising from individual claim decomposition mechanisms (Es et al., 2023).

### 4.2 Latency Overhead and Performance

A significant structural finding lies in the latency profiles: DeepEval demonstrates an approximate **7x latency overhead** per metric compared to RAGAS (mean execution of  $\sim 290s$  vs.  $\sim 40s$ ). This difference stems from DeepEval’s granular claim-decomposition approach, which generates multiple intermediate judge requests per query instance.

### 4.3 Reference-Dependent Dynamics

The reference-dependent metrics (**context\_precision** and **context\_recall**) consistently yield lower scores than reference-free metrics across both engines. This behavior aligns with system expectations, given that reference datasets consist of consolidated keyword extractions rather than complete target context answers.

## 5 Detailed Scenario Breakdown

- **Ambiguous Query Handling:** Query instances involving combined-signal edge cases revealed divergence between RAGAS’s strict entailment validation and DeepEval’s claim-level scoring (Muller et al., 2024).
- **Call Volume:** DeepEval’s higher execution time directly correlates with its increased judge-model invocation density compared to RAGAS’s single-pass approach.
- **System Robustness:** Minor JSON formatting variances and localized timeout handling were successfully managed during standard execution pipelines without process interruptions.

## 6 Conclusion

Based on low runtime overhead and seamless operational compatibility with self-hosted LLM constraints, **RAGAS** is designated as the primary evaluation framework for this pipeline. It offers significantly faster execution speeds while maintaining output fidelity aligned with larger evaluation suites.

## References

- Es, S., James, J., Espinosa-Anke, L., & Schockaert, S. (2023). Ragas: Automated Evaluation of Retrieval Augmented Generation. *arXiv*. <https://doi.org/10.48550/arxiv.2309.15217> Cited by: 1896
- Muller, S., Loison, A., Omrani, B., & Viaud, G. (2024). GroUSE: A Benchmark to Evaluate Evaluators in Grounded Question Answering. *Proceedings of the 31st International Conference on*

*Computational Linguistics* (2025), 4510–4534. <https://doi.org/10.48550/arxiv.2409.06595> Cited by: 17